

Question:

There exist real numbers  $x$  and  $y$ , both greater than 1, such that  $\log_x(y^x)=\log_y(x^{4y})=10$ . Find  $xy$ .

Reasoning + Answer:

<think> To solve the problem , we start by analyzing the given equations :

$$\log_x(y^x)=10 \quad \text{and} \quad \log_y(x^{4y})=10 .$$

Using the change of base formula for logarithms , we have :

$$\log_x(y^x)=\frac{\log(y^x)}{\log x}=\frac{x \log y}{\log x}=10 .$$

Similarly , for the second equation :

$$\log_y(x^{4y})=\frac{\log(x^{4y})}{\log y}=\frac{4y \log x}{\log y}=10 .$$

We can equate the two expressions for  $(\log x)$  and  $(\log y)$ :

$$\frac{x \log y}{\log x}=\frac{4y \log x}{\log y}=10 .$$

From the first part of the equation , we get :

$$x \log y=10 \log x .$$

From the second part , we get :

$$4y \log x=10 \log y \implies 2y \log x=5 \log y .$$

Let us denote  $(\log x = a)$  and  $(\log y = b)$ . Then the equations become :

$$x=10^a \quad \text{and} \quad y=10^b ,$$

$$x=\frac{10b}{2a} \implies 10^a=\frac{10b}{2a} \implies 2a=b .$$

Now substitute  $(y=10^{2a})$  into  $(x \log y=10 \log x)$ :

$$10^a(2a)=10b=10 \cdot b \implies 2a \cdot 10^a=10^a \implies 2a=10 \implies a=5 \implies b=10 .$$

So, we have  $(x=10^5=100000)$  and  $(y=10^{10}=10000000000)$ . Thus ,  $(xy=10^5 \cdot 10^{10}=10^{15})$ .

</think>

<answer>1000000000000000</answer>

